

Part I — Foundations of Physics · Chapter 2

Measurement, Units, and Scientific Notation

Learning to quantify the physical world with precision, consistency, and honest reporting

How long is a meter? How heavy is a kilogram? How hot is 300 kelvin? These questions seem simple, but the answers reveal something profound: every measurement in physics depends on agreed-upon standards, expressed in units, reported with appropriate precision. Before we can describe motion, force, or energy mathematically, we need a shared language of measurement. This chapter builds that language.

2.1 Why Measurement Matters in Physics

Physics is a quantitative science. The difference between describing motion as “fast” and describing it as “28 meters per second” is not merely a matter of precision; it is the difference between an impression and a testable claim. A statement that can be measured can be checked, refined, and built upon. A statement that cannot be measured remains an opinion.

Measurement connects ideas to reality. When Newton proposed that force equals mass times acceleration, that claim was not meaningful until physicists could measure force in newtons, mass in kilograms, and acceleration in meters per second squared — and verify that the equation held to within the uncertainty of their instruments. The entire structure of physics rests on this foundation: every law, every theory, every model must ultimately make predictions that can be tested by measurement.

Measurement also allows communication. A physicist in Brazil and a physicist in Japan can compare results and build on each other’s work only because they are using the same units, the same standards, and the same conventions for reporting precision. Without that shared language, science could not be cumulative.

“Measure what is measurable, and make measurable what is not so.”

— Galileo Galilei (attributed)

This chapter teaches you that shared language. By the end, you will be able to read

any measurement in physics, express it in proper scientific form, convert between units, and report it with the precision the situation demands.

2.2 Physical Quantities and Units

A physical quantity is any property of matter or energy that can be measured and expressed as a number. Length, mass, time, temperature, speed, force, and electric charge are all physical quantities. Each physical quantity has two components: a numerical value and a unit.

$$\text{Physical quantity} = \text{numerical value} \times \text{unit}$$

A measurement without a unit is physically meaningless.

For example, saying that a table is “1.8” tells us nothing. Saying that it is “1.8 meters” tells us its length precisely. The unit — the meter — is the standard against which the measurement is made. Changing the unit changes the number: the same table is 180 centimeters long, or 5.9 feet long. The physical reality does not change; only the way we express it in numbers does.

Common Mistake: Never write a measurement without its unit. In physics, “5.0” is not a length, a mass, or anything else. “5.0 m” is a length. The unit is not optional.

Physical quantities fall into two broad categories. Base quantities are the fundamental, independently defined quantities from which all others are derived. Derived quantities are combinations of base quantities. Speed, for instance, is a derived quantity: it is defined as distance divided by time, combining two base quantities.

Type	Examples	How Defined
Base quantity	Length, mass, time, temperature, current	Defined by international standard; not derived from others
Derived quantity	Speed, force, energy, pressure, density	Defined as mathematical combinations of base quantities

Table 2.1 *Base and derived quantities in physics.*

2.3 The International System of Units (SI)

Modern science uses one unified system of units: the *Système International d’Unités*, universally abbreviated SI (from the French). Adopted internationally in 1960 and

continuously refined by the International Bureau of Weights and Measures (BIPM), the SI is the agreed-upon standard for scientific measurement worldwide.

Before SI, different countries — and different fields within science — used incompatible systems. Engineers in the United States measured length in feet and force in pounds; scientists in Europe measured in centimeters and dynes; navigators used nautical miles. The resulting confusion cost real resources and, in several famous cases, caused engineering disasters. The Mars Climate Orbiter was lost in 1999 because one engineering team used metric units and another used imperial units — a \$328 million lesson in the importance of unit consistency.

Historical Note: The meter was originally defined as one ten-millionth of the distance from the North Pole to the equator. Today it is defined precisely in terms of the speed of light: one meter is the distance light travels in a vacuum in exactly $1/299,792,458$ of a second. This modern definition makes the meter independent of any physical artifact.

The SI has several important features. It is decimal: all prefixes are powers of ten, so conversions never require awkward factors like 12 (inches to feet) or 5,280 (feet to miles). It is coherent: derived units are defined as simple products or quotients of base units with no numerical factors. And it is universal: every country's national standards laboratory is traceable back to the same international definitions.

Bridge Note *In the advanced series, Volume I works entirely in SI units and introduces four-vector notation, where space and time components are combined. The consistency of SI makes that extension natural.*

2.4 Base Units and Derived Units

The SI is built on seven base units. Every other unit in physics can be expressed as a combination of these seven. You do not need to memorize the precise definitions of all seven immediately, but you should know them by name, symbol, and what they measure.

Base Quantity	SI Base Unit	Symbol	Measures
Length	meter	m	distance, displacement, size
Mass	kilogram	kg	amount of matter; resistance to acceleration
Time	second	s	duration of events; intervals

Electric current	ampere	A	flow of electric charge per second
Temperature	kelvin	K	absolute thermodynamic temperature
Amount of substance	mole	mol	number of entities (atoms, molecules)
Luminous intensity	candela	cd	power of light as perceived by the human eye

Table 2.2 *The seven SI base units. The first three — meter, kilogram, and second — are used in nearly every chapter of this book.*

Derived units are built from base units using multiplication and division. A few derived units are used so frequently that they have been given their own names and symbols to avoid writing out the full combination every time.

Derived Quantity	Derived Unit	Symbol	In Base Units
Speed / velocity	(none named)	m/s	m s^{-1}
Acceleration	(none named)	m/s^2	m s^{-2}
Force	newton	N	kg m s^{-2}
Energy / work	joule	J	$\text{kg m}^2 \text{s}^{-2}$
Power	watt	W	$\text{kg m}^2 \text{s}^{-3}$
Pressure	pascal	Pa	$\text{kg m}^{-1} \text{s}^{-2}$
Electric charge	coulomb	C	A s
Electric potential	volt	V	$\text{kg m}^2 \text{s}^{-3} \text{A}^{-1}$
Resistance	ohm	Ω	$\text{kg m}^2 \text{s}^{-3} \text{A}^{-2}$
Frequency	hertz	Hz	s^{-1}

Table 2.3 *Common derived units in SI. The base-unit expressions reveal the underlying physics.*

Why This Matters: Knowing that the newton equals $\text{kg}\cdot\text{m}/\text{s}^2$ is not just trivia. When you check whether an equation makes sense by examining its units — a process called dimensional analysis, which you will study in Chapter 3 — you will need to break

named units like newtons and joules back into their base components.

2.5 Metric Prefixes and Powers of Ten

One of the great advantages of the SI is its use of decimal prefixes. Rather than inventing a new unit for every scale — as imperial units do with inches, feet, yards, and miles — the metric system applies standard prefixes to a base unit to indicate multiples or fractions of that unit. A prefix represents multiplication by a power of ten.

Prefix	Symb ol	Power of Ten	Multiplier	Example
tera	T	10^{12}	1,000,000,000,000	1 Tm = 10^{12} m
giga	G	10^9	1,000,000,000	1 GHz = 10^9 Hz
mega	M	10^6	1,000,000	1 MW = 10^6 W
kilo	k	10^3	1,000	1 km = 1,000 m
hecto	h	10^2	100	1 hPa = 100 Pa
deca	da	10^1	10	1 dam = 10 m
(base)		10^0	1	1 m = 1 m
deci	d	10^{-1}	0.1	1 dm = 0.1 m
centi	c	10^{-2}	0.01	1 cm = 0.01 m
milli	m	10^{-3}	0.001	1 mm = 0.001 m
micro	μ	10^{-6}	0.000001	1 μ m = 10^{-6} m
nano	n	10^{-9}	0.000000001	1 nm = 10^{-9} m
pico	p	10^{-12}	0.000000000001	1 pm = 10^{-12} m
femto	f	10^{-15}	(very small)	1 fm = 10^{-15} m

Table 2.4 *SI prefixes. The most commonly used in introductory physics are kilo, centi, milli, micro, and nano.*

To convert between prefixes, you shift the decimal point or equivalently multiply or divide by the appropriate power of ten. The key skill is to always convert to base units

first when doing any calculation.

Converting with Prefixes

The method is always the same: replace the prefix with its power of ten, then simplify.

Example: Convert 45 km to meters $45 \text{ km} = 45 \times 10^3 \text{ m} = 45,000 \text{ m} = 4.5 \times 10^4 \text{ m}$

Example: Convert 320 μm to meters $320 \mu\text{m} = 320 \times 10^{-6} \text{ m} = 3.20 \times 10^{-4} \text{ m}$

Example: Convert 2.5 GHz to Hz $2.5 \text{ GHz} = 2.5 \times 10^9 \text{ Hz} = 2,500,000,000 \text{ Hz}$

Common Mistake: Prefix symbols are case-sensitive. 'm' (lowercase) is milli (10^{-3}); 'M' (uppercase) is mega (10^6). Writing 'Mm' means megameters, not millimeters. Always check capitalization.

2.6 Scientific Notation

Physics deals with quantities that span an enormous range of magnitudes. The diameter of a proton is about 0.000000000000001 meters. The distance to the nearest star beyond our Sun is about 40,000,000,000,000,000 meters. Writing these numbers in ordinary decimal form is cumbersome and error-prone. Scientific notation solves this problem.

Scientific notation expresses any number as a coefficient multiplied by a power of ten:

$$N = a \times 10^n$$

where a is a number between 1 and 10 (the coefficient), and n is an integer (the exponent).

The coefficient a must satisfy $1 \leq a < 10$. This constraint means there is exactly one non-zero digit to the left of the decimal point. The exponent n tells you how many places the decimal point has been moved, and in which direction.

The Rules of Scientific Notation

- The coefficient must be at least 1 and less than 10: write 3.2, not 32 or 0.32.
- The exponent is positive if the original number is 10 or greater.
- The exponent is negative if the original number is between 0 and 1.
- The exponent counts the number of places the decimal point moves.

Example: Convert 6,400,000 to scientific notation $6,400,000 = 6.4 \times 10^6$ (decimal moves 6 places left $\rightarrow$ positive exponent)

Example: Convert 0.00000320 to scientific notation $0.00000320 = 3.20 \times 10^{-6}$ (decimal moves 6 places right $\rightarrow$ negative exponent)

Example: Convert 1 to scientific notation $1 = 1.0 \times 10^0$ (any number to the power zero equals 1)

Arithmetic with Scientific Notation

When multiplying numbers in scientific notation, multiply the coefficients and add the exponents. When dividing, divide the coefficients and subtract the exponents. When adding or subtracting, first convert both numbers to the same power of ten.

$$(a \times 10^m) \times (b \times 10^n) = (a \times b) \times 10^{m+n}$$

$$(a \times 10^m) \div (b \times 10^n) = (a \div b) \times 10^{m-n}$$

Example: Multiply: $(3.0 \times 10^4) \times (2.0 \times 10^3) = (3.0 \times 2.0) \times 10^{4+3} = 6.0 \times 10^7$

Example: Divide: $(9.6 \times 10^8) \div (3.2 \times 10^3) = (9.6 \div 3.2) \times 10^{8-3} = 3.0 \times 10^5$

Common Mistake: After multiplying or dividing, check that your coefficient is still between 1 and 10. If you get 24×10^5 , rewrite it as 2.4×10^6 . If you get 0.6×10^{-3} , rewrite it as 6×10^{-4} .

2.7 Converting Between Standard and Scientific Notation

Converting between ordinary decimal notation and scientific notation is a mechanical skill that should become automatic. The key is to count decimal places carefully and track the sign of the exponent.

From Standard to Scientific Notation

Move the decimal point until only one non-zero digit remains to its left. Count the number of places moved. If you moved left, the exponent is positive. If you moved right, the exponent is negative.

Standard Form	Move Decimal	Exponent	Scientific Notation
93,000,000	7 places left	+7	9.3×10^7
0.00450	3 places right	-3	4.50×10^{-3}
299,792,458	8 places left	+8	2.99792458×10^8
0.0000000167	8 places right	-8	1.67×10^{-8}

1.0	no movement	0	1.0×10^0
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Table 2.5 *Worked examples of conversion to scientific notation.*

From Scientific to Standard Notation

Move the decimal point in the coefficient by the number of places given by the exponent. Positive exponent: move right (number is large). Negative exponent: move left (number is small). Fill in zeros as needed.

Example: Write 4.7×10^5 in standard form Move decimal 5 places right: $4.7 \rightarrow 470,000$

Example: Write 8.3×10^{-4} in standard form Move decimal 4 places left: $8.3 \rightarrow 0.00083$

2.8 Significant Figures

Not all digits in a number carry equal meaning. The significant figures (also called significant digits) of a measurement are the digits that carry meaningful information about the precision of the measurement. Reporting too many significant figures implies more precision than the instrument can provide; reporting too few discards information.

Rules for Counting Significant Figures

1. All non-zero digits are significant. In 34.5, there are 3 significant figures.
2. Zeros between non-zero digits are significant. In 3,007, there are 4 significant figures.
3. Leading zeros (before the first non-zero digit) are never significant. In 0.0023, there are 2 significant figures — the zeros serve only as placeholders.
4. Trailing zeros after a decimal point are significant. In 4.500, there are 4 significant figures — the trailing zeros indicate the measurement was made to the nearest thousandth.
5. Trailing zeros in a whole number without a decimal point are ambiguous. "1500" might have 2, 3, or 4 significant figures. Scientific notation removes the ambiguity: 1.5×10^3 has 2, while 1.500×10^3 has 4.

Number	Sig. Figs.	Reason
73.4	3	All non-zero digits
0.0052	2	Leading zeros not significant
1,007	4	Zero between non-zeros is significant

18.00	4	Trailing zeros after decimal are significant
1500	2 or 3 or 4?	Ambiguous without decimal or scientific notation
1.500×10^3	4	Trailing zeros in coefficient are significant
0.0200	3	Leading zeros not significant; trailing zeros are

Table 2.6 Counting significant figures. Practice until these rules are automatic.

Significant Figures in Calculations

When you multiply or divide measurements, round the result to the same number of significant figures as the measurement with the fewest significant figures.

Example: Multiplication: $4.52 \text{ m} \times 3.1 \text{ m}$ 4.52 has 3 sig. figs.; 3.1 has 2 sig. figs.
Answer rounded to 2 sig. figs.: 14 m^2 (not 14.012 m^2)

When you add or subtract measurements, round the result to the same number of decimal places as the measurement with the fewest decimal places.

Example: Addition: $12.3 \text{ m} + 0.456 \text{ m}$ 12.3 has 1 decimal place; 0.456 has 3. Answer rounded to 1 decimal place: 12.8 m (not 12.756 m)

Common Mistake: Never round in the middle of a calculation. Carry one or two extra digits through intermediate steps, then round only the final answer. Rounding too early causes accumulated rounding error.

Bridge Note In the advanced series, significant figures give way to formal uncertainty propagation — a statistical treatment of how errors compound through calculation. The habits built here make that transition straightforward.

2.9 Precision vs. Accuracy

Precision and accuracy are often used interchangeably in everyday speech, but in physics they have distinct meanings that are essential to understanding what a measurement tells you.

Accuracy refers to how close a measurement is to the true value of the quantity being measured. An accurate measurement is correct — it reflects reality. Precision refers to how reproducible a measurement is — how close repeated measurements are to each other, regardless of whether they are close to the true value.

Case	Accurate?	Precise?	Description
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Darts clustered near bullseye	Yes	Yes	Ideal: low random and systematic error
Darts clustered away from bullseye	No	Yes	Systematic error (bias): consistently wrong
Darts scattered near bullseye	Yes	No	High random error: correct on average, inconsistent
Darts scattered away from bullseye	No	No	Both systematic and random error: worst case

Table 2.7 *The four combinations of accuracy and precision, illustrated by the dart-board analogy.*

A measurement can be precise without being accurate. A miscalibrated balance might give the same reading every time (high precision) while being systematically off by 5 grams (low accuracy). Conversely, a measuring technique might on average give the right answer (high accuracy) while individual measurements scatter widely (low precision).

Key Distinction: Precision is limited by random error — the unpredictable variations inherent in any measurement process. Accuracy is limited by systematic error — consistent biases caused by faulty instruments, improper technique, or incorrect assumptions.

In experimental physics, you want both. The goal is to reduce random error (by repeating measurements and averaging) and eliminate systematic error (by calibrating instruments and checking assumptions).

2.10 Measurement Error and Instrument Limits

Every measuring instrument has a limit of resolution — the smallest difference it can detect. A ruler marked in millimeters cannot reliably measure lengths smaller than about half a millimeter. A stopwatch that displays hundredths of a second cannot be read to thousandths. The instrument's resolution sets a lower bound on the uncertainty of a direct measurement.

Reading Analog Instruments

For analog instruments (rulers, graduated cylinders, analog meters), the convention is to estimate to one decimal place beyond the smallest marked division. If a ruler has millimeter markings, you read to the nearest 0.1 mm. This estimated digit is uncertain, but it still carries information.

Example: Reading a ruler A ruler marked in millimeters shows an object between the

4.3 cm and 4.4 cm marks, appearing closer to 4.4 cm. A good reading is 4.37 cm $\pm$ 0.05 cm. The 7 is estimated; the $\pm$ 0.05 cm expresses the uncertainty.

Reading Digital Instruments

Digital instruments display a fixed number of digits. The last displayed digit is typically uncertain by ± 1 . A digital balance reading 12.34 g implies an uncertainty of about ± 0.01 g — the last place shown.

Expressing Uncertainty

A measurement is written as a value plus or minus its uncertainty:

$$\text{measurement} = \text{best estimate} \pm \text{uncertainty}$$

For example: $L = 3.52 \text{ m} \pm 0.01 \text{ m}$. This tells the reader that the true value is believed to lie between 3.51 m and 3.53 m. The uncertainty is not a sign of weakness; it is honest information about the reliability of the measurement.

Percent Error and Percent Uncertainty

When a true or accepted value is known, percent error measures how far off a measurement is:

$$\text{Percent error} = \frac{|\text{measured} - \text{accepted}|}{\text{accepted}} \times 100\%$$

Percent uncertainty expresses uncertainty as a fraction of the measurement:

$$\text{Percent uncertainty} = \frac{(\text{uncertainty} \div \text{measured value})}{1} \times 100\%$$

Example: Percent uncertainty If a mass is measured as 48.5 g $\pm$ 0.5 g, the percent uncertainty is $(0.5 \div 48.5) \times 100\% \approx 1.0\%$.

2.11 Estimation and Order of Magnitude

Not every physical problem requires a precise measurement. Sometimes the most useful question is: roughly how big is this? Estimation — the art of making reasonable guesses without exact data — is one of the most important skills in physics.

An order of magnitude is the nearest power of ten that represents the size of a quantity. The mass of a proton is about 10^{-27} kg. The mass of the Earth is about 10^{25} kg. These are order-of-magnitude estimates; the exact values differ by factors of two or three, but the powers of ten capture the essential scale.

The Fermi Method

The physicist Enrico Fermi was famous for making accurate estimates of seemingly impossible quantities from limited information. His approach — now called Fermi estimation — breaks a problem into smaller sub-questions that can each be estimated from common knowledge, then combines them.

Classic Fermi Problem: How many piano tuners are there in Chicago? Estimate the population (about 3 million), the fraction of households with a piano ($\approx 1/20$), the number of tunings per year per piano (≈ 1), and the number of tunings a tuner can do per day (about 4) times working days (about 250). Result: roughly $3,000,000 \div 20 \div (4 \times 250) \approx 150$ tuners. The actual answer is around 100–200.

The value of the Fermi method is not the exact answer but the process: identifying the key variables, making defensible estimates for each, and assembling them into a result that is almost certainly within a factor of ten of reality. A factor of ten error in physics is often called “getting the order of magnitude right,” and it is frequently all that is needed to decide whether a proposed mechanism is plausible.

Order-of-Magnitude Estimates in Practice

- Always carry units through your estimate — they guide you toward the right combination of quantities.
- Round aggressively: use 10 for “about a dozen,” use 10^3 for “a few thousand.”
- Check your answer against physical intuition: if your estimate gives the mass of a car as 10 kg, something went wrong.
- An estimate that is off by a factor of two or three is usually acceptable. An estimate off by a factor of 1,000 is not.

2.12 Recording and Reporting Measurements Properly

The conventions for recording and reporting measurements exist for one reason: to communicate the information content of a measurement as clearly and honestly as possible. A reader of your lab report should be able to reproduce your measurement and understand exactly what precision it has.

The Standard Format

Every measurement you record in physics should include:

6. The numerical value (best estimate).
7. The unit (always, no exceptions).
8. The uncertainty, expressed as $\pm$ with appropriate significant figures.
9. The number of significant figures consistent with the uncertainty.

Example: $m = 125.3 \text{ g} \pm 0.1 \text{ g}$

The uncertainty ($\pm 0.1 \text{ g}$) justifies reporting to the tenths place (4 significant figures).

Consistency Between Value and Uncertainty

The uncertainty must be expressed to the same number of decimal places as the measurement. You would never write $12.3456 \text{ m} \pm 0.1 \text{ m}$ — this implies the last four decimal places are meaningless. You would write $12.3 \text{ m} \pm 0.1 \text{ m}$, or $12.35 \text{ m} \pm 0.05 \text{ m}$ if your instrument supports that precision.

Common Mistake: Never report a measurement as -12.3456 m from a ruler marked in centimeters. The instrument cannot support that precision. Write $-12.3 \text{ cm} \pm 0.1 \text{ cm}$ or $-12.35 \text{ cm} \pm 0.05 \text{ cm}$, then convert if needed.

Laboratory Notebook Standards

- Record measurements in ink, not pencil. If you make an error, cross it out with a single line so the original value is still readable.
- Record measurements at the time you make them, not from memory.
- Always note the instrument you used and its stated precision or resolution.
- Include units immediately after every numerical entry.
- If a measurement seems wrong, note it as 'suspect' and repeat. Never erase or destroy original data.

Bridge Note *In the advanced series, uncertainty is treated statistically using standard deviation and error propagation formulas. The clear, honest recording habits built here are the foundation for that quantitative treatment in Volumes I through V.*

Chapter Summary

This chapter established the mathematical language physicists use to quantify the physical world. Every measurement in physics must state a numerical value, a unit, and an uncertainty. The SI provides a universal, coherent, decimal-based system of units built on seven base quantities.

Key ideas from Chapter 2:

- Every physical quantity has a numerical value and a unit. A number without a unit is not a physical measurement.
- The SI (Système International) is the universal system of measurement in science, built on seven base units.

- All other units in physics are derived units — combinations of the seven base units.
 - Metric prefixes (kilo, centi, milli, micro, nano, etc.) are powers of ten applied to base units, making conversion strictly decimal.
 - Scientific notation expresses any number as a coefficient between 1 and 10 multiplied by a power of ten.
 - Significant figures communicate the precision of a measurement. Rules for sig. figs. in calculations: multiply/divide → fewest sig. figs.; add/subtract → fewest decimal places.
 - Accuracy measures closeness to the true value; precision measures reproducibility. Both are needed; neither alone is sufficient.
 - Every measurement has uncertainty. Report measurements as value $\pm$ uncertainty, with matching significant figures and decimal places.
 - Order-of-magnitude estimation is a core physics skill: round aggressively, carry units, and check plausibility.
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Key Terms

physical quantity: Any property of matter or energy that can be measured and expressed as a numerical value with a unit.

unit: A standard reference amount against which a physical quantity is measured.

SI (Système International): The internationally agreed-upon system of units built on seven base units, used universally in science.

base unit: One of the seven fundamental SI units from which all other units are derived: meter, kilogram, second, ampere, kelvin, mole, candela.

derived unit: A unit formed by combining base units through multiplication or division; examples include newton, joule, and watt.

metric prefix: A prefix applied to an SI base or derived unit to indicate multiplication by a power of ten; examples include kilo (10^3), centi (10^{-2}), and nano (10^{-9}).

scientific notation: A way of expressing any number as a coefficient between 1 and 10 multiplied by a power of ten: $a \times 10^n$.

significant figures: The digits in a measurement that carry meaningful information about its precision, counted according to specific rules.

accuracy: How close a measurement is to the true or accepted value of the quantity

being measured.

precision: How reproducible a measurement is; how close repeated measurements are to one another.

random error: Unpredictable, non-repeating variation in measurements caused by inherent limitations of the measurement process.

systematic error: Consistent, repeatable deviation from the true value caused by a miscalibrated instrument, faulty technique, or flawed assumption.

uncertainty: The range of values within which the true value of a measurement is believed to lie; expressed as $\pm$ a value.

percent error: A measure of accuracy: $|\text{measured} - \text{accepted}| \div \text{accepted} \times 100\%$.

percent uncertainty: Uncertainty expressed as a percentage of the measured value: $(\text{uncertainty} \div \text{measurement}) \times 100\%$.

order of magnitude: The nearest power of ten representing the size of a quantity; a rough measure of scale.

resolution: The smallest difference an instrument can reliably detect or display.

Fermi estimation: A problem-solving method in which a complex quantity is estimated by breaking it into sub-quantities that can each be reasonably guessed.

Concept Review Questions

Answer the following in complete sentences. No calculation required.

10. Why is a number without a unit meaningless in physics? Give two examples to illustrate your answer.
11. What are the seven SI base units? List each one, its symbol, and what physical quantity it measures.
12. Explain the difference between a base unit and a derived unit. Give one example of each.
13. Why does the metric system make unit conversion easier than the imperial system (feet, inches, pounds)?
14. What does the exponent in scientific notation tell you, and what does its sign indicate?
15. Why is the rule that the coefficient in scientific notation must be between 1 and 10 useful?

16. State the rules for counting significant figures. Apply each rule to one example of your own.
 17. Explain the difference between accuracy and precision in your own words. Describe a situation in which a measurement could be precise but not accurate.
 18. What is a systematic error? Give one realistic example from a laboratory setting and explain how you might detect and correct it.
 19. What does it mean to say a measurement has an uncertainty of ± 0.05 m? What does this tell you about the true value?
 20. Why is order-of-magnitude estimation a useful skill even when precise calculations are possible?
 21. A student records a length as "42 centimeters" in her lab notebook. What is missing from this record? What should she write instead?
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Measurement Practice Problems

Show all work. Include units in every step. Round final answers to the correct number of significant figures.

Unit Conversions

22. Convert 5.50 km to meters. Express your answer in scientific notation.
23. Convert 250 mA (milliamperes) to amperes.
24. Convert 3.2×10^{-4} m to micrometers (μm).
25. Convert 1 light-year (approximately 9.461×10^{15} m) to kilometers.
26. A car travels 60.0 miles per hour. Convert this speed to meters per second. (1 mile = 1609 m)
27. The diameter of a human hair is about 70 μm . Express this in (a) meters, (b) millimeters, (c) nanometers.

Derived Unit Calculations

28. A force of 35 N acts on an object with mass 5.0 kg. What is the acceleration? State the derived units of your answer and confirm they match m/s^2 .
29. A box has dimensions 0.40 m $\times$ 0.25 m $\times$ 0.15 m and a mass of 3.6 kg. Calculate its density in kg/m^3 .
30. An engine produces 4,500 J of work in 15 s. Calculate the power in watts. Verify the unit: 1 W = 1 J/s.

Uncertainty and Percent Error

31. A student measures the length of a pendulum three times: 82.3 cm, 82.5 cm, 82.4 cm. (a) Find the average. (b) Find the range (max – min). (c) Report the measurement with uncertainty.
32. The accepted value for the acceleration due to gravity is 9.81 m/s^2 . A student measures 9.65 m/s^2 . Calculate the percent error.
33. A mass is measured as $45.2 \text{ g} \pm 0.2 \text{ g}$. What is the percent uncertainty?
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Scientific Notation Exercises**Part A: Convert to Scientific Notation**

34. 0.000 000 167
35. 5,972,000,000,000,000,000,000 (mass of Earth in kg)
36. 0.00350
37. 186,000 (approximate speed of light in miles per second)
38. 0.1
39. 1,000,000,000

Part B: Convert to Standard Notation

40. 6.022×10^{23}
41. 1.67×10^{-27} (mass of a proton in kg)
42. 3.00×10^8
43. 9.11×10^{-31}

Part C: Arithmetic in Scientific Notation

44. $(2.5 \times 10^3) \times (4.0 \times 10^5)$
45. $(8.4 \times 10^{12}) \div (2.1 \times 10^4)$
46. $(6.0 \times 10^{-3}) \times (5.0 \times 10^{-4})$
47. $(3.0 \times 10^8) \div (6.0 \times 10^3)$
48. $(4.0 \times 10^3) + (2.5 \times 10^2)$ [convert to same exponent first]
49. $(7.2 \times 10^{-4}) - (3.0 \times 10^{-5})$
-

Significant Figures Exercises

Part A: Count Significant Figures

State the number of significant figures in each measurement and explain your reasoning.

50. 0.00340 m

51. 1,050 kg

52. 2.000×10^3 s

53. 10,000 g (no decimal point shown)

54. 0.10800 mL

55. 3.14159

Part B: Round Correctly

Round each number to the number of significant figures indicated.

56. 7.8546 to 3 significant figures

57. 0.004397 to 2 significant figures

58. 12,850 to 3 significant figures (use scientific notation)

59. 99.97 to 3 significant figures

Part C: Apply Sig. Fig. Rules in Calculations

60. $(4.25 \text{ m}) \times (1.7 \text{ m}) = ?$ (state the answer with correct sig. figs. and unit)

61. $(100.5 \text{ g}) - (2.35 \text{ g}) = ?$

62. $(6.3 \text{ m/s}) \times (4.00 \text{ s}) = ?$

63. $(9.6 \times 10^3 \text{ J}) \div (3.00 \times 10^2 \text{ s}) = ?$

64. $(5.280 \text{ km}) + (342 \text{ m converted to km}) = ?$

Lab / Measurement Activity

Measurement Reliability Lab — Comparing Instruments

Purpose

To practice taking measurements with instruments of different precision, to observe the effect of instrument resolution on significant figures, to calculate average values and uncertainties, and to distinguish random from systematic error.

Materials

- A small cylindrical object (coin, bottle cap, or similar)
- A ruler with millimeter markings
- A digital caliper (if available)
- A triple-beam balance or digital scale
- A stopwatch (phone timer acceptable)

Procedure

65. Using the millimeter ruler, measure the diameter of the cylindrical object five times, repositioning the object each time. Record each measurement in mm to the nearest 0.1 mm (estimated digit).
66. If a caliper is available, repeat with the caliper. Record to the nearest 0.1 mm or 0.01 mm as the caliper allows.
67. Measure the mass of the object five times using the balance. Record each reading to the precision of the instrument.
68. Time how long it takes you to tap the desk ten times, five separate trials. Record each time in seconds.

Data Analysis

69. For each set of five measurements, calculate: (a) the mean (average), (b) the range (max – min), (c) the uncertainty as half the range, (d) the percent uncertainty.
70. Report each quantity in the format: value $\pm$ uncertainty (unit).
71. Compare the ruler and caliper results (if both were used). Which instrument gave smaller uncertainty? Explain why in one sentence.
72. Identify one possible source of random error and one possible source of systematic error in your diameter measurement.

Reflection Questions

73. If you repeated the diameter measurement 50 times instead of 5, would your precision improve? Your accuracy? Explain.
74. Suppose the caliper's jaws were slightly bent, causing it to consistently read 0.3 mm too large. How would this affect accuracy? Precision? How might you detect this error?
75. Express your average diameter in scientific notation. How many significant figures are justified by your uncertainty?

Chapter 2 Checkpoint

Before moving on to Chapter 3, you should be able to do the following without looking at your notes. If you cannot, review the relevant section.

I can...	See Section
Explain why every measurement requires both a number and a unit	2.1–2.2
Name the seven SI base units and their symbols	2.3–2.4
Distinguish base units from derived units and give examples of each	2.4
Apply metric prefixes to convert between scales (km, cm, mm, μm , nm)	2.5
Write any number in correct scientific notation (coefficient between 1 and 10)	2.6
Convert between standard decimal form and scientific notation	2.7
Multiply and divide numbers in scientific notation	2.6–2.7
Count the significant figures in any measurement	2.8
Apply sig. fig. rules when multiplying, dividing, adding, and subtracting	2.8
Explain the difference between accuracy and precision with an example	2.9
Distinguish random error from systematic error	2.10
Express a measurement with its uncertainty: value $\pm$ uncertainty (unit)	2.10–2.12
Calculate percent error and percent uncertainty	2.10
Make and defend an order-of-magnitude estimate using the Fermi method	2.11

Record a measurement in proper lab-notebook format	2.12
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Continue to Chapter 3: Dimensional Analysis →